

Precautionary Electrification*

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Abstract

I study the distributional effects of gas price volatility on households' energy adoption (extensive margin) and consumption (intensive margin). Two facts motivate the analysis. First, residential gas prices have historically been more volatile than electricity prices and are projected to remain so. Second, poorer households disproportionately rely on electricity, while richer households consume relatively more gas. I develop a heterogeneous-agent model of residential energy demand in which households with non-homothetic preferences choose between gas, whose price is stochastic, and electricity, whose price is stable. Calibrated to U.S. data, the model replicates observed patterns of fuel use and responses to price shocks. I find that higher volatility induces poorer households to shift toward electricity, while richer households remain exposed. These results highlight volatility as a driver of electrification and have substantial implications for the distribution of welfare, and energy policy.

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1 Introduction

Volatile energy prices have emerged as a defining feature of the global economy in recent years. Russia’s invasion of Ukraine in 2021, following the COVID-19 crisis, exposed households and policy-makers to sharp and unpredictable increases in energy costs. Beyond geopolitical shocks, structural changes such as the shale boom and the globalization of natural gas markets have contributed to persistent volatility in gas prices, in contrast with the relative stability of electricity prices. These developments underscore a central question: how do households adjust their energy choices in response to differences in the volatility of energy prices?

The residential sector provides a particularly salient context. In the United States, households account for roughly one-third of final energy consumption (EIA, 2024). Because energy expenditures represent a meaningful share of household budgets, especially among the poor, price fluctuations have both significant distributional consequences and aggregate implications. Importantly, households are heterogeneous in both their energy use and their vulnerability to shocks? Higher-income households are more likely to rely on natural gas for heating, while lower-income households disproportionately use electricity. This heterogeneity in fuel choice implies that energy price shocks – and their volatility – may have distributional consequences that extend beyond standard expenditure shares.

This paper studies how households’ energy consumption and fuel choices respond to volatility in energy prices. I begin by documenting two motivating facts. First, over the past three decades, natural gas prices have displayed greater volatility than electricity prices (Apergis, Bowden, and Payne, 2015, Regnier, 2007, Chan and Grant, 2016), Wiggins and Etienne, 2017. The 2021-22 energy crisis amplified this divergence, with gas prices experiencing extreme and unpredictable spikes. Second, households’ reliance on electricity versus gas is systematically correlated with income: using panel data from the PSID, I show that a 1% percent increase in income is associated with a 9 percentage point lower probability of using

electricity as the primary heating fuel. Together, these facts suggest a potential link between income heterogeneity, fuel choice, and exposure to energy price volatility.

To analyze this link, I develop a heterogeneous-agent model of residential energy consumption in which households choose both the type and amount of energy they consume. Households face non-homothetic preferences, consistent with the empirical fact that the share of energy in total expenditure declines with income (e.g. Känzig, 2023). Gas prices follow a stochastic process with exogenous shocks, while electricity prices remain constant. Switching fuels is possible but entails financial and utility costs. In this environment, poorer households – who devote a larger share of expenditures to energy – are more exposed to gas price volatility and thus face stronger precautionary incentives to adopt electricity.

I calibrate the model to match empirical patterns in U.S. household energy use. The model reproduces key features of the data, including the positive relationship between income and energy consumption, the declining energy expenditure share with income, and the observed income gradient in fuel choice. It also replicates the differential response of gas and electricity consumption to gas price shocks. I then use the calibrated model to study how an increase in gas price volatility affects household behavior across the income distribution. The results indicate that higher volatility induces a shift from gas to electricity, particularly among poorer households. Richer households, by contrast, remain exposed to volatility because they are less likely to switch.

These findings have two key implications. First, heterogeneity in households' energy use may be shaped not only by differences in upfront costs or product quality, but also by the volatility of energy prices. In particular, persistent gas price volatility may have accelerated the shift of poorer households toward electricity. Second, policies aimed at cushioning households from short-term price shocks – such as the subsidies implemented during the 2021-22 energy crisis – may inadvertently slow the electrification of the residential sector, with consequences for the green transition. More broadly, the results highlight the importance of energy price

volatility as a driver of household behavior, with implications for both welfare and climate policy.

Related literature. This paper relates to four strands of the literature: the effects of energy price shocks, the role of uncertainty in household decision-making, household fuel choice, and climate macroeconomics.

A large literature studies the effects of energy price changes on households and the macroeconomy. Within this literature, one line of research estimates households' price elasticities of energy demand (e.g., Alberini, Gans, and Velez-Lopez, 2011, Alberini and Filippini, 2011, Alberini, Khymych, and Ščasný, 2019, Miller and Alberini, 2016, Gelman et al., 2023). Another line studies the aggregate effects of energy price shocks (e.g. Pindyck and Rotemberg, 1983, Sweeney, 1984), often exploiting the exogeneity of oil shocks (e.g. Kilian, 2008, Kilian and Lewis, 2011, Edelstein and Kilian, 2009, Kilian, 2014, Kilian and Vigfusson, 2011, Gagliardone and Gertler, 2023). More recently, work has examined the heterogeneous distributional effects of energy price shocks across households (e.g. Känzig, 2023; Auclert et al., 2023; Bobasu, Dobrew, and Repele, 2025; Aguilar and Fuentes-Albero, 2025). I contribute to this literature by focusing not on the level of prices but on their volatility, and by distinguishing between electricity and gas consumption, thereby uncovering an important channel of heterogeneity in exposure to shocks.

This paper is also related to work on how aggregate uncertainty affects household behavior. Coibion et al. (2024) show that higher macroeconomic uncertainty reduces household spending on nondurables, while Kumar, Gorodnichenko, and Coibion (2023) study the effects of uncertainty on firms' decisions. Alberini, Ščasný, et al. (2023) provide direct evidence that households form expectations about energy prices and that they are especially uncertain about electricity and gas prices. In Europe, Cevik and Zhao (2025) measures the spillovers from electricity price volatility across countries. My contribution is to show that households' expectations about energy price volatility influence not only consumption level but also fuel

choice, with long-run implications for the composition of residential energy demand.

A third strand examines the determinants of household’ adoption of different fuels. Davis (2011) and Davis (2025) studies the role of energy prices in households’ adoption of electricity; Rapson and Bushnell (2024) analyzes the diffusion of electric vehicles; and Allcott, Kwon, and Snyder (2024) highlights distributional aspects of electrification. I extent this literature by emphasizing volatility rather than levels of energy prices as a determinant of fuel choice. In my framework, households anticipate that gas prices are more volatile than electricity prices, and poorer households – who devote a larger share of expenditures to energy – are more likely to switch to electricity in response to that risk.

Finally, this paper contributes to the growing literature on macroeconomics and climate policy. Foundational work in this area began with Integrated Assessment Models (Nordhaus, 1993), and subsequent contributions have studied optimal carbon taxes (Golosov et al., 2014), fiscal instruments for mitigation (Barrage, 2020), climate policy uncertainty (Fried, Novan, and Peterman, 2021), and macro-financial aspects of the green transition (Gilchrist, Martinez, and Rickard, 2024). More recent research embeds energy into heterogeneous-agent models, building on Aiyagari (1994). Examples include Fried (2024) on temperature shocks, Känzig (2023) on the distributional effects of carbon pricing, and Krusell and Smith Jr (2022) on IAMs with heterogeneous agents. My paper adds to this literature by modeling energy consumption along both the extensive (fuel type) and intensive (quantity) margins, incorporating non-homothetic preferences that generate heterogeneous exposure to energy price volatility. The results suggest that volatility itself may be an underexplored channel accelerating residential electrification.

Roadmap. The rest of this paper is organized as follows. Section 2 presents the empirical motivation and evidence on household fuel choice using the PSID. Section 3 develops and calibrates the heterogeneous-agent model of energy consumption. Section 4 uses the model to study the effects of gas price volatility across the income distribution. Section 5 concludes

with implications for energy policy and the green transition.

2 Motivating Facts

This section presents two empirical facts that motivate the analysis. First, residential gas prices have been far more volatile than electricity prices over the past quarter century, reflecting differences in how the two markets are structured and how they transmit supply shocks. Second, households are heterogeneous in their choice of energy source: richer households disproportionately rely on gas, while poorer households rely more heavily on electricity, largely due to differences in heating technology. Taken together, these facts imply that income groups face systematically different exposure to energy price risk, providing the central motivation for the model developed below.

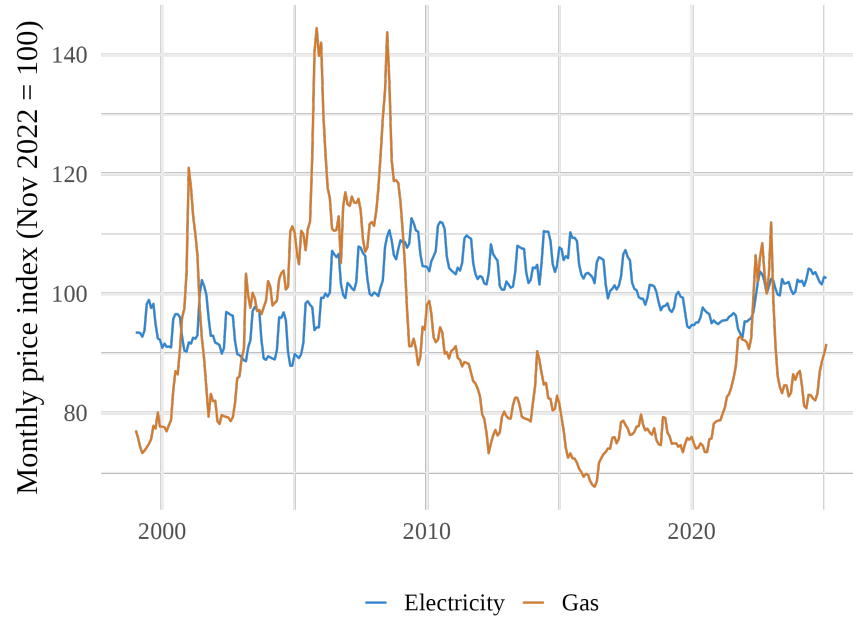
2.1 The relative volatility of residential gas prices

Figure 1 plots monthly U.S. residential price indices for electricity and natural gas from January 1999 to February 2025, deflated to November 2022 dollars using the CPI. To make volatility more visible, I present the raw series without de-seasonalization, so that both seasonal fluctuations and longer-term swings are apparent. The contrast between the two series is striking: residential gas prices are considerably more volatile than electricity prices, a pattern that holds throughout the sample.

Several distinct periods can be identified in the gas series. Between 1999 and 2009, volatility was extremely high, with a coefficient of variation of 18%.¹ From 2009 until the eve of the COVID-19 pandemic, volatility fell markedly, with the coefficient of variation dropping to 8.8%. Since 2021, volatility has risen again to 9.3%, during a period shaped by the global energy crisis following Russia’s invasion of Ukraine. The existence of such shifts in volatility

¹The coefficient of variation is defined as the standard deviation divided by the mean, which allows comparison of relative volatility across periods with different average price levels.

Figure 1: Monthly US residential electricity and gas price indices, 1999-2025



is consistent with the findings of Hailemariam and Smyth (2019), who estimate a structural VAR with stochastic volatility and identify significant regime breaks in gas price uncertainty, closely matching the patterns visible in the raw data.

Electricity prices, in contrast, have remained remarkably smooth. While they display predictable seasonal patterns, the overall level of volatility has been consistently low: the coefficient of variation of monthly electricity prices was 6.3% between 1999 and 2009, fell to 4.4% between 2009 and 2021, and declined further to 3.3% since 2021. In other words, if anything, electricity prices have become less volatile over time, in sharp contrast to the fluctuations observed in gas.

Why is gas so much more volatile than electricity? The key lies in how the two markets are organized. Natural gas prices are determined in wholesale commodity markets, increasingly integrated with international trade since the early 2000s, when gas markets were deregulated (Apergis, Bowden, and Payne, 2015, Wiggins and Etienne, 2017, Hailemariam and Smyth, 2019). As a result, they are highly sensitive to exogenous supply shocks: the shale gas

boom in the late 1990s and early 2000s generated large downward swings in prices, while more recently, geopolitical disruptions such as the war in Ukraine triggered large spikes. Because gas is traded internationally and storage is costly, short-run shocks to supply or transportation constraints translate quickly into end-use prices.

Electricity, by contrast, is consumed locally and delivered through highly regulated utilities (U.S. Energy Information Administration, 2023). Even though natural gas is a key input into electricity generation, residential electricity tariffs are buffered from wholesale price volatility by regulation, long-term contracts, cross-subsidization, and the mix of other generation sources (coal, nuclear, renewables). In effect, the institutional and technological structure of the electricity sector absorbs much of the volatility present in the underlying gas market. This explains why households face far more stable electricity prices even when natural gas, upstream, is subject to dramatic swings.

Taken together, the evidence highlights a robust stylized fact: U.S. residential gas prices are volatile, subject to sharp regime shifts driven by exogenous supply shocks, whereas electricity prices are comparatively smooth, reflecting both regulation and diversified production. This asymmetry provides the central motivation for the model: when households choose between gas and electricity, they face not only different price levels, but fundamentally different degrees of price risk.

2.2 Household income and energy choice

A second key motivating fact is that households are heterogeneous in the type of energy they consume, with poorer households relying disproportionately on electricity and richer households on natural gas. This pattern matters because, as shown above, gas prices are substantially more volatile than electricity prices. Heterogeneity in energy adoption therefore translates directly into heterogeneity in exposure to price risk.

Data. I document these patterns using the Panel Study of Income Dynamics (PSID), a nationally representative longitudinal survey collected biennially since 1968. I focus on waves from 1999 to 2023, when data on households’ main heating fuel and energy expenditures are consistently available. The PSID provides rich information on household characteristics, including demographics, housing attributes, and detailed expenditures. Most relevant for this study, the survey records the household’s primary heating fuel and annual expenditures on electricity and natural gas.

To account for climate heterogeneity, I merge the restricted-use PSID geocodes with daily temperature data from the National Oceanic and Atmospheric Administration (NOAA), constructing annual heating and cooling degree days at the census-tract level (base 65°F/18°C). Following standard PSID practice, I restrict the sample to households that report electricity or gas as their main heating source and exclude immigrant refresher sub-samples. I also drop implausible observations with annual energy expenditures below \$0 or above \$40,000 (2022 USD), or annual total expenditures below \$2,000. After these restrictions, the final sample contains 45,589 household-year observations for 9,598 unique households, averaging roughly 3,500 observations per year. All monetary values are deflated to 2022 USD using the CPI (all items), and longitudinal survey weights are applied.

Descriptive patterns. Figure 2 plots the share of households reporting electricity and gas as their main heating fuel between 1999 and 2023. The share of electricity users has risen steadily over time, while the share of gas users has declined. Figure 3 shows that this aggregate trend masks substantial heterogeneity: electrification has been concentrated among poorer households. By 2021, more than half of bottom-quintile households (Q1) reported electricity as their primary heating fuel, compared to fewer than one-third of top-quintile households (Q5).

These differences in technology choice translate into systematic differences in energy expenditure patterns. Figure 4 reports average annual energy expenditures by income quintile and

heating type. As expected, households with gas heating devote most of their spending to gas, while electricity-heated households spend almost exclusively on electricity. Higher-income households spend more in absolute terms, but energy accounts for a declining share of total expenditures, a clear sign of non-homotheticity in energy demand.²

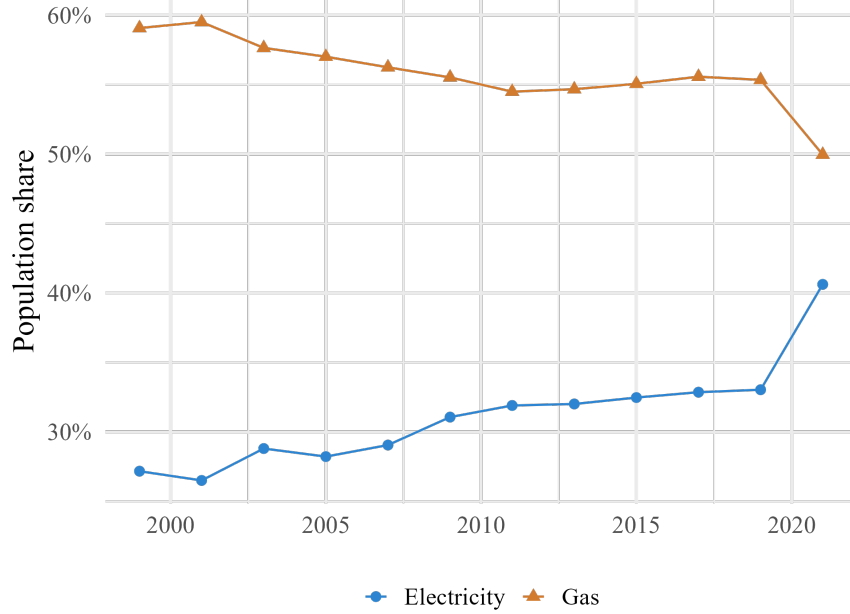


Figure 2: Evolution of Main Heating Fuel Over Time

Regression evidence. To quantify the income gradient in energy adoption, I estimate a linear probability model of electricity use:

$$\text{hasElectricity}_{ist} = \beta_0 + \beta_1 \log(\text{income}_{it}) + X_{it} + \gamma_t + \gamma_s + \varepsilon_{ist},$$

where $\text{hasElectricity}_{ist}$ indicates whether household i in state s and year t relies on electricity as its main heating fuel. Income is measured as four-year average household income, X_{it} includes controls for climate (heating and cooling degree days), housing characteristics, homeownership status, family size, and head's age. I also include year and state fixed effects (γ_t, γ_s) to absorb time and region-specific factors, including permanent differences in energy

²I describe non-homothetic expenditure in energy further in Appendix X.

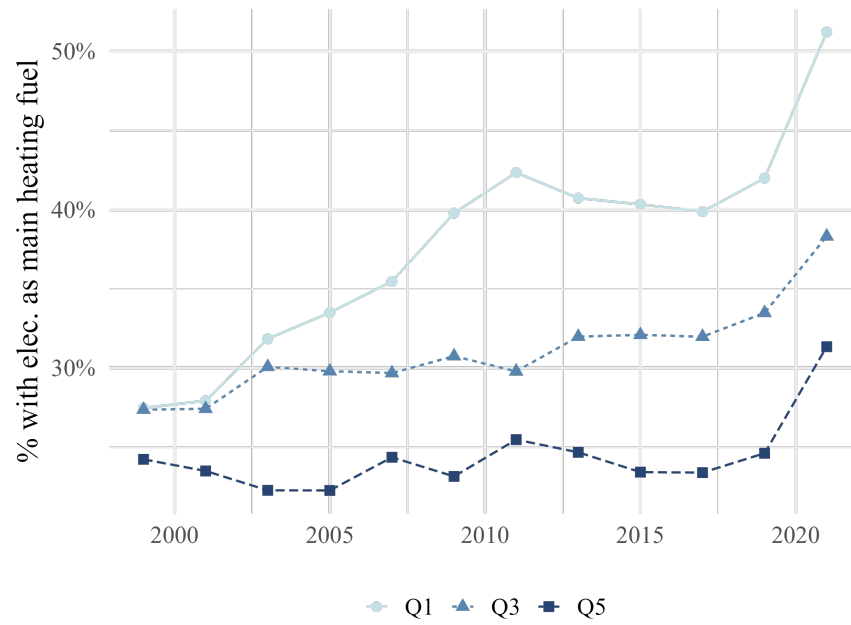


Figure 3: Electrification share, by income quintile

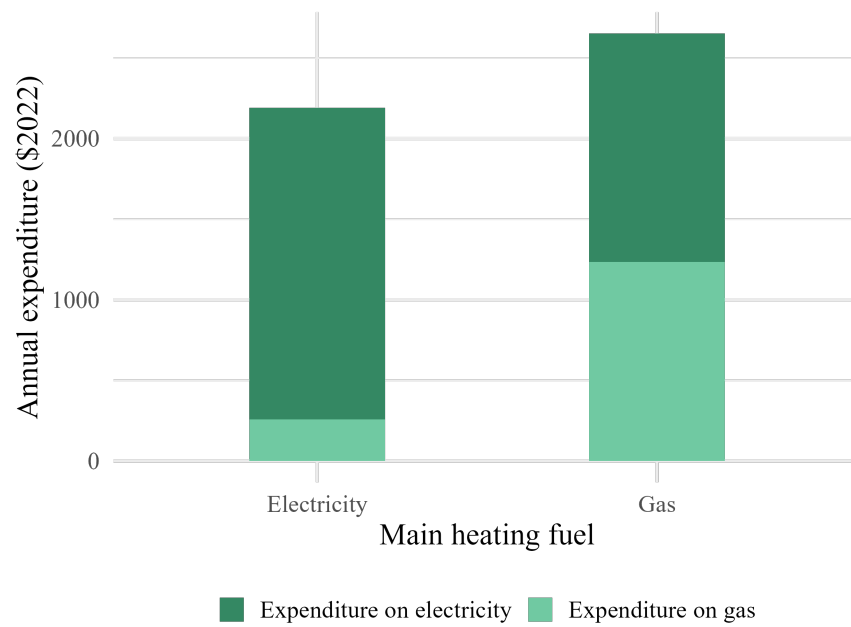


Figure 4: Energy Expenditure by Main Heating Fuel and Income Quintile

prices.

Table 1 summarizes the regression results across three specifications: (1) unadjusted; (2) with covariates; and (3) with covariates and fixed effects. Across all models, the coefficient β_1 is negative and statistically significant, confirming that poorer households are systematically more likely to rely on electricity. In the fully saturated model with controls and fixed effects, a 1% increase in income is associated with a 0.09 percentage point decrease in the likelihood of using electricity.

Table 1: Income and Electricity adoption

	(1) Baseline	(2) + controls	(3) + year/state FE
log income	−0.284*** (0.002)	−0.192*** (0.003)	−0.089*** (0.021)
Controls	No	Yes	Yes
Year FE	No	No	Yes
State FE	No	No	Yes
Observations	38,105	33,705	33,663
R^2	0.01	0.134	0.228

Notes: Coefficients are from logit regressions. Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The dependent variable is a binary indicator for electricity as main heating fuel. Controls include heating and cooling degree-days (base 65°F/18°C), age (squared), house tenure, family size (squared) and house type.

This results implies that energy choices are strongly stratified by income. Poorer households are disproportionately reliant on electricity, while richer households are more likely to adopt gas. Combined with the fact that energy expenditures decline as a share of income while gas prices are more volatile, this heterogeneity implies that energy price shocks have unequal effects across the income distribution. Poor households, by relying on the less volatile energy source, effectively insure themselves against risk; richer households, by consuming more gas, are more directly exposed to volatility. This income–energy link is central to the quantitative model that follows, where households choose both how much energy to consume and which

energy source to adopt in the presence of gas price uncertainty.

3 Model

In this section, I first introduce the quantitative model used to study how volatility in gas prices affects households' energy adoption and consumption decisions. The model is a heterogeneous-agent Huggett-type economy with incomplete markets, idiosyncratic income risk, and an extensive margin of energy choice between gas and electricity. Preferences are non-homothetic. I then show the model implications for households' behavior, the calibration, and the model fit.

3.1 Environment

Time is discrete and infinite. A unit mass of infinitely lived households receives a stochastic endowment z which follows the following process:

$$\log z' = \rho_z \log z + \sigma_z \varepsilon'_z, \quad \varepsilon'_z \sim \mathcal{N}(0, 1), \quad (1)$$

where ρ_z measures persistence and σ_z the volatility of idiosyncratic shocks.

Households consume two goods: a non-energy composite c_c and an energy good c_k . Energy can be supplied by one of two technologies, $k \in \{\ell, g\}$, where ℓ denotes electricity and g gas. A household owns one technology at a time but may switch at a cost.

Non-homothetic preferences. Preferences are defined over a composite index c , which aggregates c_c and c_k according to the non-homothetic CES structure of Comin, Lashkari, and Mestieri (2021), with two goods:

$$(1 - \theta)^{\frac{1}{\mu}} \left(\frac{c_c}{c} \right)^{\frac{\mu-1}{\mu}} + \theta^{\frac{1}{\mu}} \left(\frac{c_k}{c^\eta} \right)^{\frac{\mu-1}{\mu}} = 1. \quad (2)$$

Here, μ governs the elasticity of substitution between energy and non-energy goods. The share parameter θ determines the relative weight of energy in the composite index c . Finally, η is the non-homotheticity parameter for energy; note that since there are only two goods in this specification, I set the non-energy composite good c_c as the baseline good, and as such normalized its non-homotheticity parameter to 1. Therefore, when $\eta < 1$, energy is “closer to a necessity”, implying that the energy expenditure share declines with income.

This specification is strictly non-homothetic: marginal budget shares change smoothly with income, allowing the model to match Engel curve patterns across the entire distribution. By contrast, Stone–Geary preferences impose a hard subsistence level and are asymptotically homothetic, making them ill-suited to capture energy demand patterns in the data.

Lifetime utility is:

$$U = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}, \quad (3)$$

with coefficient of relative risk aversion σ and discount factor β .

Energy prices. Gas prices q_g are stochastic, while electricity prices q_ℓ are constant:

$$q_g(\gamma) = \gamma \bar{q}_g, \quad \log \gamma' = \rho_\gamma \log \gamma + \sigma_\gamma \varepsilon'_\gamma, \quad \varepsilon'_\gamma \sim \mathbb{N}(0, 1), \quad (4)$$

$$q_\ell = \bar{q}_\ell. \quad (5)$$

The AR(1) shock γ captures volatility in gas prices, with persistence ρ_γ and innovation variance σ_γ . Electricity prices are stable, consistent with the fact that residential electricity rates in the U.S. are often regulated and insulated from wholesale fluctuations, as discussed in Section 2.

I treat energy prices as exogenous, i.e. the model is solved in partial equilibrium. This choice is motivated by both empirical and conceptual considerations. Empirically, U.S. households

represent only about 15% of total natural gas demand, compared with roughly 30% for the industrial sector (EIA, 2023). As a result, changes in residential consumption are too small to materially influence aggregate gas prices. Similarly, electricity prices faced by end-use consumers reflect regulated tariffs and infrastructure costs more than short-run demand fluctuations. Conceptually, the focus of the paper is on households’ heterogeneous responses to volatility, not on general-equilibrium feedbacks through energy markets. Modeling prices as exogenous therefore isolates the distributional and welfare consequences of volatility from supply-side dynamics.

This simplifying assumption comes with trade-offs. It rules out feedback channels whereby shifts in household energy demand could affect aggregate energy markets or investment decisions by utilities. Such channels are particularly relevant for policy counterfactuals, for example when studying large-scale electrification of heating or the interaction between gas and electricity in power generation. I return to this point in the conclusion, where I discuss extensions of the model that endogenize prices. For the purposes of this paper, however, partial equilibrium is the appropriate environment: it captures the household-side heterogeneity that motivates the analysis while keeping the model tractable and tightly linked to the empirical facts.

Energy choice. At the beginning of each period, households choose whether to continue using its current technology k or to switch to the alternative $k' \neq k$. Switching is costly along two distinct dimensions. First, it requires the household to pay a fixed financial cost f . This cost does not depend on the household’s income or on which direction the switch takes place (from gas to electricity or from electricity to gas). Economically, it can be interpreted as the upfront outlay associated with replacing heating equipment, upgrading infrastructure, or undertaking other installation expenses.

Second, switching entails a non-pecuniary cost that reflects households’ heterogeneous tastes for different fuels. I model this by assuming that when evaluating the alternative technology

k' , the household draws a taste shock $\zeta_{k'}$ from a Type I Extreme Value (Gumbel) distribution with location parameter $\mu_{k'}$ and scale parameter λ . The location parameters $\mu_{k'}$ shift average preferences toward one technology or the other, while the scale parameter λ governs the degree of dispersion in idiosyncratic tastes across households. A larger λ implies more heterogeneity in household preferences, and hence less responsiveness of aggregate switching behavior to differences in relative prices.

Together, these two costs capture both the financial and the subjective barriers to fuel switching. The financial cost f ensures that households do not switch technologies too frequently in response to transitory shocks, while the taste shocks introduce a smooth, probabilistic choice structure rather than a knife-edge decision rule. This formulation follows the standard random utility framework of Adda and Cooper (2000) and allows the model to match the observed distribution of fuel choices across households.

Budget constraint and composite expenditure. Given composite c , energy type k , and gas shock γ , the minimum expenditure required to attain c is:

$$E(c, k; \gamma) = \left[(1 - \theta) c^{(1-\mu)} + \theta c^{(1-\mu)\eta} q_k(\gamma)^{1-\mu} \right]^{\frac{1}{1-\mu}}, \quad (6)$$

where $q_k(\gamma) = q_\ell$ if $k = \ell$, and $q_k(\gamma) = q_g(\gamma)$ if $k = g$. Appendix 1 shows the derivations to get to this expression.

Intuitively, the term in brackets balances the relative importance of non-energy and energy goods (calibrated with $(1 - \theta)$ and θ), their relative necessity (calibrated with η), and the price of the energy technology chosen.

The budget constraint and borrowing constraints are:

$$E(c, k; \gamma) + a' = (1 + r)a + z, \quad a' \geq -\phi, \quad (7)$$

where a' is next-period assets and ϕ is the borrowing limit.

Recursive problem. Let $V(a, z, k; \gamma)$ denote the value function at the start of a period. The household's state consists in its assets a , its idiosyncratic productivity z , its current energy type $k \in \{g, \ell\}$ and the aggregate gas shock γ . The household compares the value of staying with k or switching to k' :

$$V(a, z, k; \gamma) = \max_{\{\text{keep}, \text{switch}\}} \left\{ \nu(a, z, k; \gamma), \nu(a - f/(1 + r), z, k'; \gamma) + \zeta_{k'} \right\}, \quad (8)$$

where $\nu(a, z, k; \gamma)$ is the value conditional on owning energy type k and is defined as:

$$\nu(a, z, k; \gamma) = \max_{c, a' \geq -\phi} \frac{c^{1-\sigma} - 1}{1 - \sigma} + \beta \mathbb{E}[V(a', z', k; \gamma')], \quad (9)$$

subject to the budget constraint (7) and shock processes for z (1) and γ (4).

Equilibrium. The equilibrium concept in this environment is partial equilibrium. Since energy prices are taken as exogenous, the model does not impose market clearing in energy markets. Instead, equilibrium requires internal consistency between households' decision rules, the exogenous processes governing income and gas price volatility, and the resulting distribution of household states.

Formally, an equilibrium consists of (i) value functions and associated policy functions for consumption, savings, and fuel switching that solve the household's dynamic programming problem; (ii) expectations over future income and gas prices that are consistent with the AR(1) laws of motion specified in equations (1) and (4); and (iii) a stationary distribution $\Psi(a, z, k; \gamma)$ over the individual state space that is invariant under the optimal decision rules and the stochastic processes for $(z; \gamma)$.

Because the model is solved in partial equilibrium, the equilibrium object is entirely determined by household behavior: given exogenous energy prices, each household optimally

chooses whether to switch fuels and how much to consume and save, and the aggregate distribution Ψ reflects the accumulation of these decisions. In this sense, equilibrium reduces to the internally consistent mapping from household states and shocks into decision rules and, ultimately, into an invariant distribution.

3.2 Optimal decision rules and aggregates

I now look at the inner workings of the model and its implications for households' energy-related decisions. Given a state $(a, z, k; \gamma)$, households first compute the value from switching energy type and from not switching energy type. This involves solving the consumption-savings problem in both cases. They then compare both values given the taste shock drawn this period.

Consumption-savings problem. The consumption-savings problem is described in (9). I first show households' savings decisions given a consumption index c , and then show the consumption decision for c_c and c_k , given a consumption index c .

The Euler equation in this problem is:

$$\frac{c_t^{-\sigma}}{\partial E(c_t, k_t; \gamma_t)/\partial c_t} \geq \beta(1+r) \mathbb{E}_t \left[\frac{c_{t+1}^{-\sigma}}{\partial E(c_{t+1}, k_{t+1}; \gamma_{t+1})/\partial c_{t+1}} \right], \quad (10)$$

with equality whenever the borrowing constraint does not bind (i.e., $a' > -\phi$). The derivations are shown in Appendix 1. Equation (10) says that when $a' > -\phi$, the household equates the marginal utility of one more dollar of expenditure today (left-hand side) to the discounted expected marginal utility value of a dollar saved (right-hand side). The non-homothetic structure enters through the partial derivative of the expenditure function E with respect to the consumption index c (this is the endogenous price index, see Appendix 1): because energy is closer to a necessity, this rate of change falls less rapidly with c than in a homothetic CES, and depends on which technology k is in place and on the realized

gas shock γ . Fuel switching affects the Euler equation only through this and on the resource constraint, via the fixed cost f in periods when switching occurs.

The expenditure on c_c and c_k is found as the values minimizing the expenditure on c , and are given by:

$$c_c = (1 - \theta) E(c, k; \gamma)^\mu c \quad (11)$$

$$c_k = \theta \left[\frac{E(c, k; \gamma)}{q_k(\gamma)} \right]^\mu c^\eta, \quad (12)$$

Of particular interest is the expression for energy consumption, (12). A higher value of $q_k(\gamma)$ is associated with a lower energy consumption – this is the intensive channel through which energy prices affect energy consumption.

The decision to switch. Households face idiosyncratic taste shocks $\zeta_k \sim \text{Gumbel}(\lambda, \mu_k)$, which affect the utility from switching to each energy source k . Given those shocks, the probability that a household switches from energy type k to $k' \neq k$, conditional on its asset level a , idiosyncratic income state z , and the aggregate gas price shock γ , is given by

$$\Pr(\text{switch} \mid a, z, k; \gamma) = \frac{1}{1 + \exp \left[\frac{\nu(a, z, k; \gamma) - \nu(a - f, z, k'; \gamma) - \mu_{k'}}{\lambda} \right]}.$$

This switching decision depends fundamentally on the difference in the value of staying with the current energy type versus switching:

$$\Delta\nu(a, z, k; \gamma) \equiv \nu(a, z, k; \gamma) - [\nu(a - f, z, k'; \gamma) + \mu_{k'}]$$

The probability of switching is decreasing in $\Delta\nu$: if $\Delta\nu(a, z, k; \gamma)$ increases, the relative value of the current energy source rises, making switching less attractive; if $\Delta\nu(a, z, k; \gamma)$ decreases, switching becomes more likely. This is shown in Figure 5, panel (a). As the gas shock increases, both the value from keeping gas and the value from switching decrease,

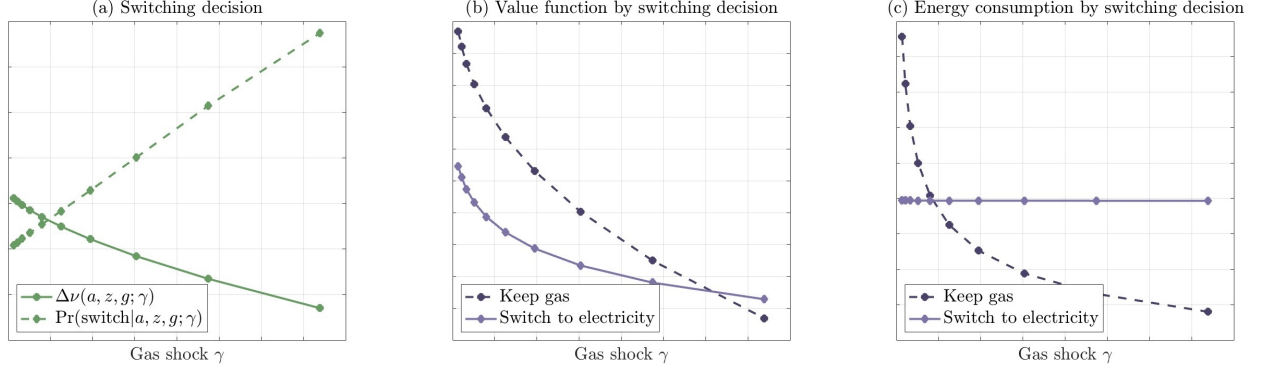


Figure 5: Switching decision for a household currently owning gas, i.e. $k = g$

though at varying rates (Figure 5, panel (b)), thereby leading to an increase in the probability to switch to electricity. For a household currently owning gas, i.e. with $k = g$, the value from keeping gas decreases faster than the value from switching to electricity because a higher gas shock γ leads to a decrease in energy consumption (Figure 5, panel (c)).

The model also implies that poorer households are more likely to adopt electricity as a response to gas price shocks than richer households. This is due to non-homothetic preferences. Because of non-homothetic preferences, poorer households spend a larger share of total expenditure on energy. In effect, they must spend a minimum amount on energy (subsistence level) in order to start enjoying their consumption of energy. Therefore, an increase in the price of gas (higher gas shock) has a greater effect on poorer households than on richer households, and this effect is due to non-homothetic preferences. This is shown in Figure 6. Panel (a) shows the difference in value, $\Delta\nu(a, z, g; \gamma)$ as a function of the gas shock γ , for three levels of income. The difference is smaller for richer households (dark blue), which leads to a lower probability to switch (panel (b)). The difference is smaller for richer households because their energy expenditure share if they keep gas (panel (c)) increases less fast for those households than poorer households. If they decide to switch, their energy expenditure share does not depend on the gas shock, by design. Therefore, non-homothetic preferences induce poorer households to respond to the shock.

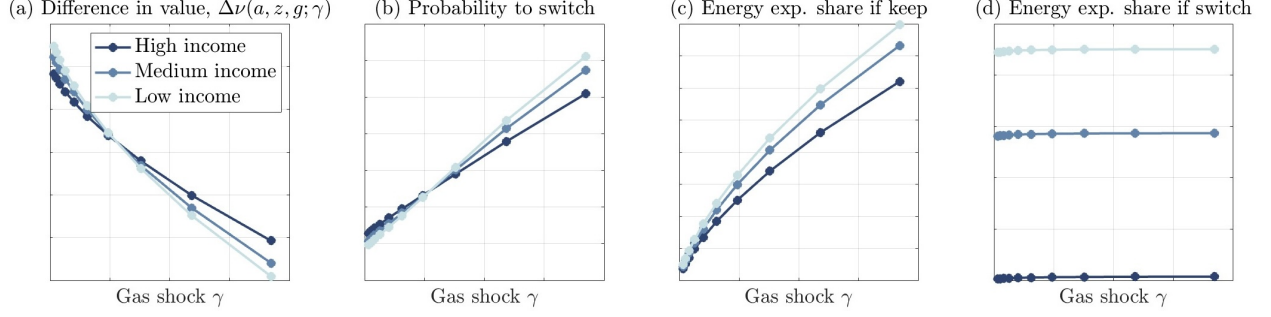


Figure 6: Caption

Aggregates. Let households follow their optimal policy functions for non-energy consumption $c_c(a, z, k; \gamma)$, energy consumption $c_k(a, z, k; \gamma)$ with $k \in \{g, \ell\}$, asset accumulation $a'(a, z, k; \gamma)$, and switching decisions. Aggregate objects are then defined by integrating these policies over the stationary distribution $\Psi(a, z, k; \gamma)$ of households across assets, income, energy type, and gas shocks.

Gas consumption. Aggregate demand for gas, C_g , consists of two groups: households who already own gas and choose to keep it, and households who initially own electricity but decide to switch. Formally,

$$C_g = \int c_k(a, z, g; \gamma) \Pr(\text{keep}|a, z, g; \gamma) d\Psi + \int c_k(a, z, g; \gamma) \Pr(\text{switch}|a, z, \ell; \gamma) d\Psi. \quad (13)$$

The first term corresponds to “stayers,” while the second captures “switchers into gas.”

Electricity consumption. Analogously, total electricity demand is

$$C_\ell = \int c_k(a, z, \ell; \gamma) \Pr(\text{keep}|a, z, \ell; \gamma) d\Psi + \int c_k(a, z, \ell; \gamma) \Pr(\text{switch}|a, z, g; \gamma) d\Psi, \quad (14)$$

combining electricity stayers with switchers out of gas.

Energy expenditure. Given prices, aggregate energy expenditure E is the dollar value of total

energy consumed:

$$E = \int [q_g(\gamma) c_k(a, z, g; \gamma) + \bar{q}_\ell c_k(a, z, \ell; \gamma)] d\Psi. \quad (15)$$

Equivalently, in time-series form it can be written as the long-run average

$$E = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=0}^T [q_g(\gamma_t) C_{g,t} + \bar{q}_\ell C_{\ell,t}], \quad (16)$$

where $C_{g,t}$ and $C_{\ell,t}$ denote period- t aggregate quantities.

Non-energy expenditure. The aggregate of non-energy consumption is simply

$$C_c = \int c_c(a, z, k; \gamma) d\Psi. \quad (17)$$

Energy share. Finally, the aggregate energy expenditure share in steady state is defined as

$$s_E = \frac{E}{E + C_c}. \quad (18)$$

These aggregates play a central role in calibration. First, the relative price of gas $\bar{q}_g$ and the electricity price $\bar{q}_\ell$ jointly pin down the ratio of aggregate gas to electricity demand in steady state. Second, the overall level of energy prices determines the aggregate energy expenditure share s_E , which is the key moment capturing non-homotheticity in the data: as income rises, households reduce the fraction of their budget devoted to energy. Thus, energy price levels discipline the mapping from the micro preference parameters (θ, η, μ) into aggregate expenditure shares, while relative prices discipline the calibration of the switching margin between electricity and gas. Put differently, the cross-sectional distribution of fuel types is sensitive to relative prices, whereas the Engel curve for energy expenditure is disciplined by the overall level of prices. This structure provides the empirical leverage to identify the key

parameters of the model.

3.3 Quantification

I calibrate the model to U.S. data around the 2019–2021 period, treating this as the relevant steady state. The model is annual, and the calibration proceeds in two steps. First, I assign values to a subset of parameters, fixing them at standard values or borrowing from the literature. Second, conditional on these assignments, I estimate the remaining parameters by simulated method of moments (SMM), choosing them so that the model reproduces a set of targeted empirical moments. The full parameterization is summarized in Table 2.

Table 2: Model parameterization

I. Assigned parameters			II. Estimated parameters		
σ	1	ccra coefficient	β	0.956	discount factor
r	0.03	interest rate	η	0.040	energy non-homotheticity
ϕ	0	borrowing constraint	$\bar{q}_g$	0.036	unit cost of gas
μ	0.275	energy/non-energy elast. subst.	$\bar{q}_\ell$	0.030	unit cost of electricity
ρ_z	0.920	income shock persistence	μ_g	0.662	switching taste shock, elec
σ_z	0.285	income shock s.d.	μ_ℓ	−0.693	switching taste shock, gas
ρ_γ	0.8	gas shock persistence	f	0.067	switching fixed cost
σ_γ	0.4	gas shock s.d.	λ	0.044	switching taste shock scale
θ	0.07	energy utility share			

Assigned parameters. Standard values are imposed where available. Preferences are logarithmic ($\sigma = 1$), the annual interest rate is set to 3%, and households cannot borrow ($\phi = 0$). Energy and non-energy goods are complements, with an elasticity of substitution $\mu = 0.275$ following Känzig (2023). Income shocks follow an AR(1) process with persistence 0.92 and standard deviation 0.285, values consistent with the HANK literature. Gas shocks are modeled with persistence 0.8 and standard deviation 0.4; in counterfactuals, I vary σ_γ to study the effect of increased volatility. Finally, the utility weight on energy θ is set equal to the observed aggregate energy expenditure share of 7% (Comin, Lashkari, and Mestieri, 2021).

Estimated parameters. The remaining parameters are identified using simulated method of moments. The discount factor $\beta = 0.956$ matches the aggregate wealth-to-income ratio from the 2019 Survey of Consumer Finances. The non-homotheticity parameter $\eta = 0.040$ is disciplined by energy expenditure shares: it ensures that low-income households allocate a larger share of their budget to energy, while reproducing the observed ratio of energy shares between the top and bottom quintiles. Unit costs of gas and electricity are pinned down by the relative expenditures of households with different heating fuels, as well as the aggregate energy-to-non-energy expenditure ratio. I target the electricity-to-gas expenditure ratio, focusing solely on the expenditure for space heating, as per the EIA estimations from the 2023 RECS.

Switching behavior disciplines the remaining parameters. The location parameters of the taste-shock distribution (μ_g, μ_ℓ) , the fixed financial cost f , and the scale parameter λ are jointly identified by the observed fraction of gas users, the aggregate switching probability, and the relative likelihood of gas adoption between the top and bottom quintiles. This ensures that the model matches both the average prevalence of each fuel type and the income gradient in technology choice.

Table 3: Targeted moments

Targeted moment	Data	Model
Wealth-to-income ratio	5.53	5.53
Energy/non-energy expenditure ratio	0.07	0.06
Electricity/gas expenditure ratio	0.89	0.87
Energy expenditure share, Q1	0.10	0.10
Energy expenditure share, Q5	0.04	0.04
Probability of switching	0.15	0.15
Share with gas	0.59	0.60
Share with gas, Q5/Q1	1.30	1.30

3.4 Model fit

Steady-state. Figure 7 compares the model to the data by income quintile along three dimensions: total energy expenditure, the energy expenditure share, and the share of households using gas. The model matches all three patterns closely. In particular, it replicates the decline in energy expenditure shares with income and the strong income gradient in gas adoption, two of the motivating facts emphasized earlier.

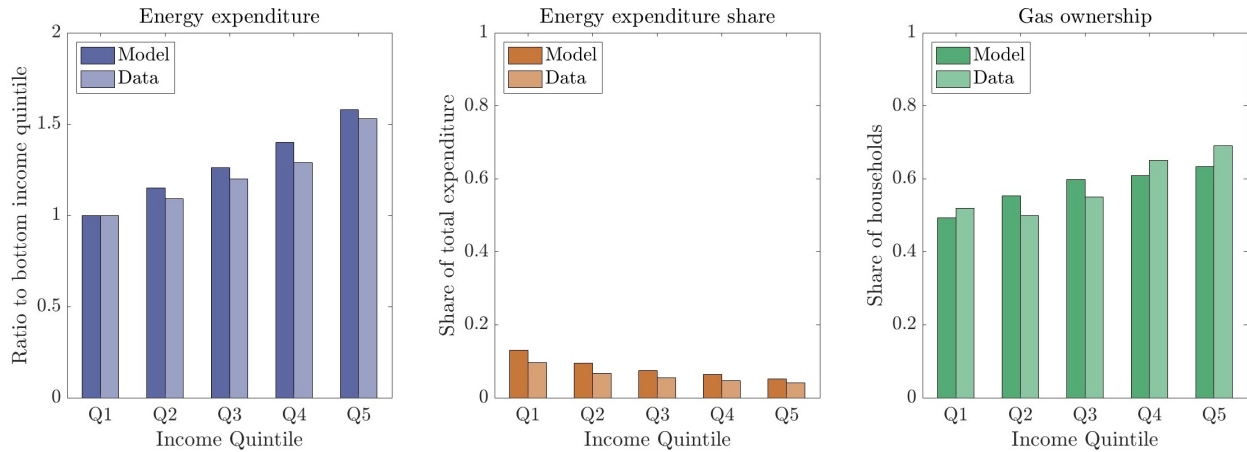


Figure 7: Model Fit by Income Quintile: Energy Use, Share, and Energy Type

Dynamics. To assess the model’s dynamic performance, I compute its implied price elasticities of gas and electricity demand. I simulate 1,000 households over 10,000 periods and regress log consumption on log gas prices, controlling for log income, log assets and the households’ binary switching decision. The coefficient on gas prices gives the own-price elasticity of gas demand, while the corresponding response of electricity demand provides the cross-price elasticity. Table X reports the results alongside empirical estimates from Alberini, Gans, and Velez-Lopez (2011). Their estimates place the own-price elasticity of gas demand between -0.693 and -0.566 , and the cross-price elasticity of electricity demand (with respect to gas prices) between 0.102 and 0.139 . The model produces an own-price elasticity of gas demand of -0.61 , which lies inside this range. However, the model implied substitution from gas to electricity is stronger than is observed empirically. This most likely

Table 4: Gas-price elasticities: model vs. empirical estimates

Own-price elasticity of gas	Cross-price elasticity of electricity
I. Model-implied	
−0.61 (0.00)	0.80 (0.00)
II. Empirical estimates (Alberini et al. 2011)	
[−0.69, −0.57]	[0.10, 0.14]

stems from the fact that, in the model, electricity prices are left unaffected by gas price shocks. As such, households respond to gas price increases by shifting more aggressively toward electricity. In reality, electricity prices are partly tied to fuel costs and regulatory adjustments, which dampens this substitution and makes the empirical cross-price elasticity smaller. Nonetheless, the model reproduces the correct direction of substitution: electricity demand rises in response to higher gas prices.

4 The Volatility Channel of Electrification

This section quantifies how a permanent rise in the volatility of gas prices reshapes household energy choices and welfare. Using the calibrated model, I show that higher volatility pushes households toward electricity and that this “precautionary electrification” is stronger among poorer households. I then discuss policy implications.

4.1 A permanent increase in gas price volatility

I consider an unanticipated, permanent 10% increase in gas price volatility: σ_γ rises from 0.40 to 0.44. The transition is computed in the standard way. I first solve for the new stationary equilibrium under the higher volatility. I then compute the perfect-foresight transition from the initial steady state to the new one by backward induction on a finite horizon, using the policy functions that are time-indexed along the path. Finally, I simulate the economy

forward from the initial steady state to obtain aggregate and distributional dynamics.

Aggregate effects. Figure 8 shows the transition of aggregate energy and non-energy consumption. Both rise on impact, with non-energy consumption increasing more than energy consumption. The modest movement in aggregate energy masks an internal reallocation: electricity consumption rises while gas consumption falls (Figure 9). The decline in gas use reflects a fall in the share of households adopting gas as their main technology (Figure 10). In short, higher volatility induces a shift toward the *less risky* technology even when relative *average* prices are unchanged, consistent with a precautionary motive operating at the extensive margin.

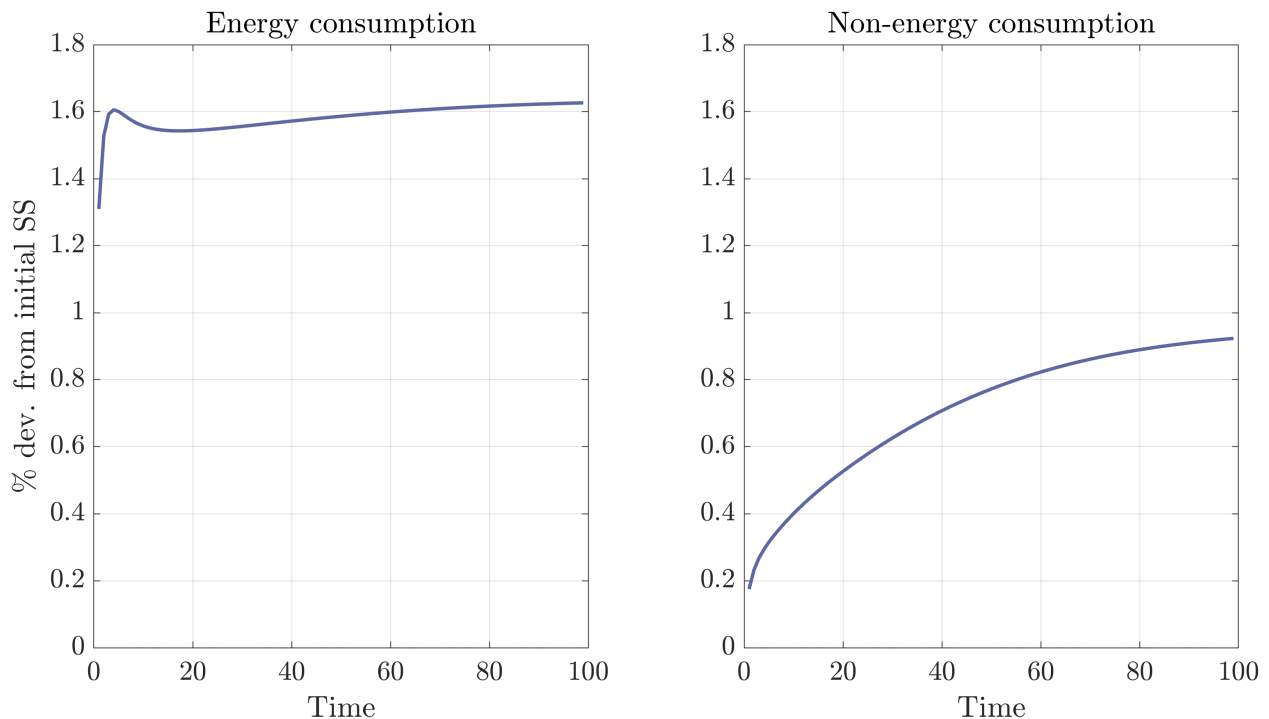


Figure 8: Transition paths of aggregate energy and non-energy consumption following a 10% increase in gas price volatility

Distributional effects. Figure 11 decomposes responses by income group. Total energy consumption rises on impact for all groups, but the increase is less persistent for the bottom quintile. For non-energy consumption, higher-income households expand spending, whereas

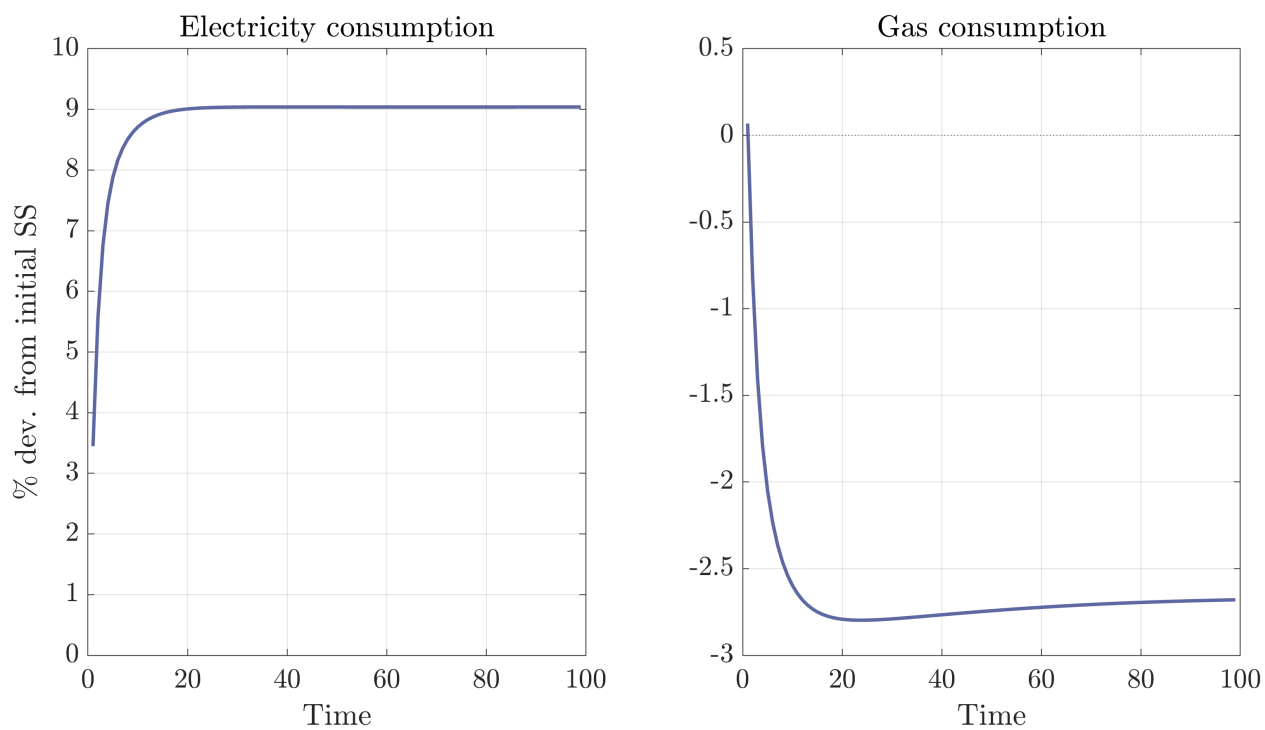


Figure 9: Transition paths of electricity and gas consumption

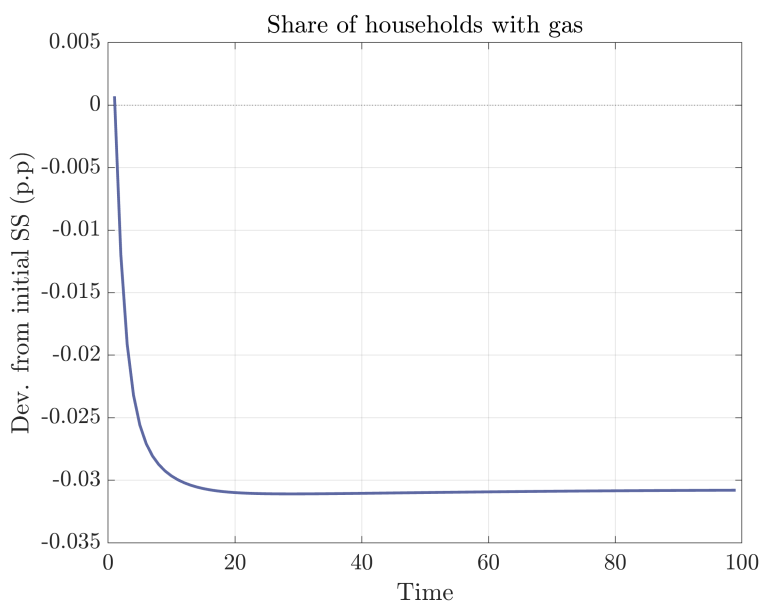


Figure 10: Transition of the share of households using gas as main heating technology

lower-income households initially contract it, reflecting tighter budget constraints and the cost of switching.

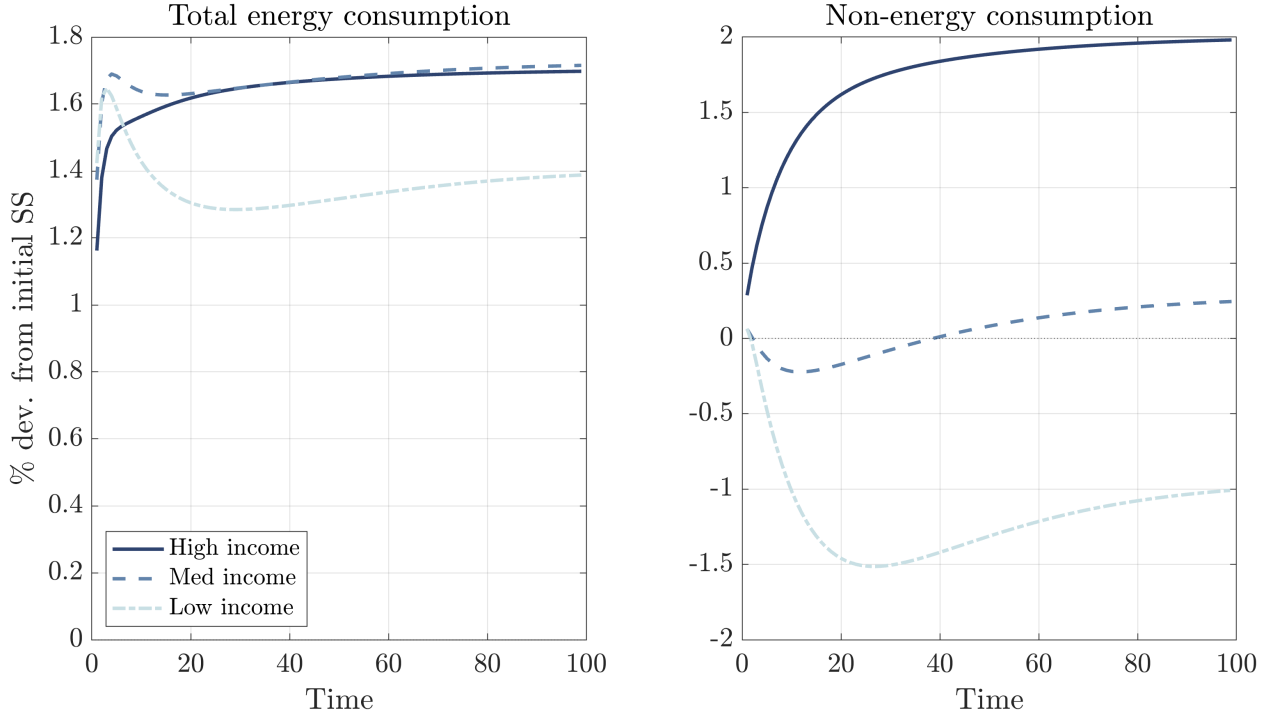


Figure 11: Energy and non-energy consumption by income group

Figure 12 clarifies the mechanism by fuel. Electricity use rises across the distribution, but the increase is larger for richer households in levels. Gas consumption falls everywhere, with a larger decline at the top simply because high-income households begin with higher baseline gas use. The extensive-margin response is more pronounced among the poor: Figure 13 shows a sharper drop in gas adoption in the bottom quintiles. Intuitively, with non-homothetic preferences, poorer households devote a higher budget share to energy, making them more exposed to gas price risk and thus more responsive to volatility.

Welfare effects. Welfare is evaluated in consumption-equivalent units. For each household i , I compute the certainty-equivalent consumption ω_i that delivers lifetime utility V_i under

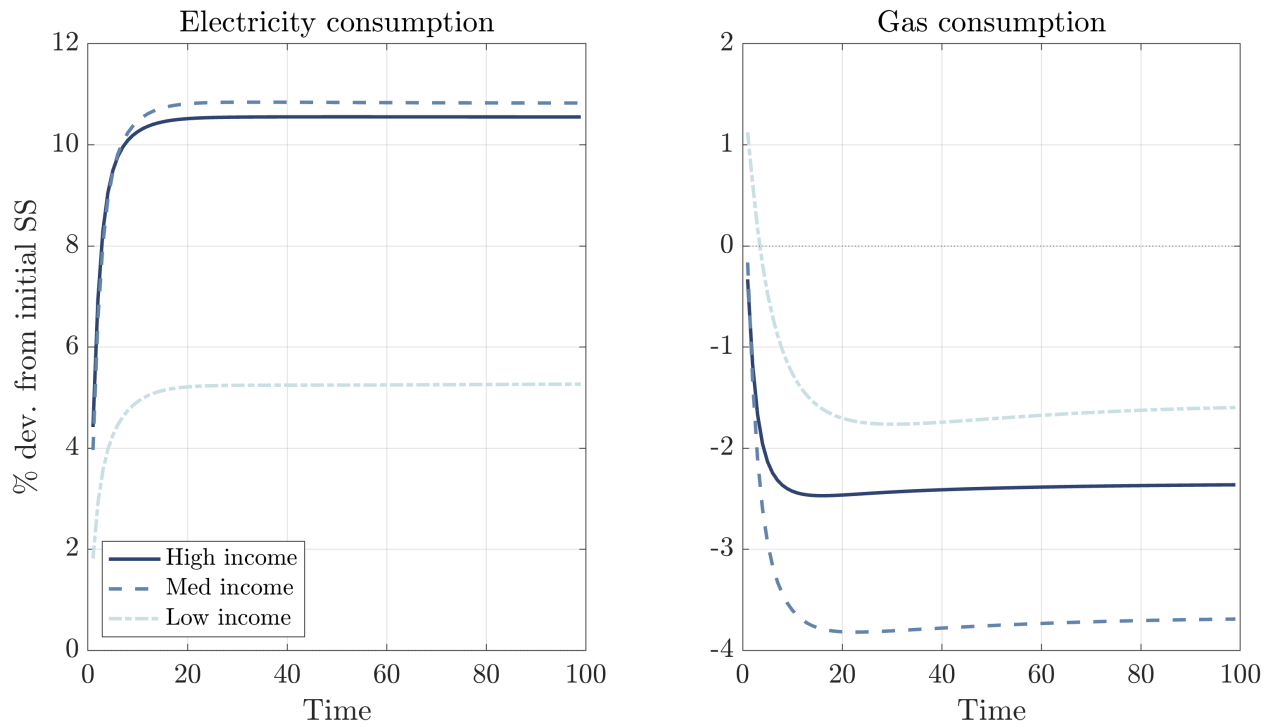


Figure 12: Electricity and gas consumption by income group

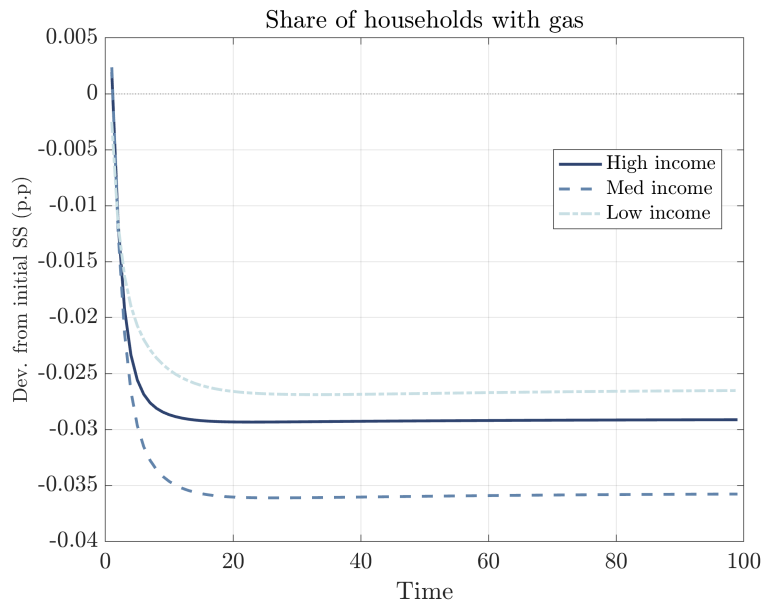


Figure 13: Share using gas as main fuel, by income group

a constant $c = \omega_i$ each period, i.e.

$$V_i = \sum_{t=0}^{\infty} \beta^t u(\omega_i) \Rightarrow \omega_i = u^{-1}[(1 - \beta) V_i].$$

Figure 14 reports the change in consumption-equivalent welfare by income quintile. On average, the welfare effect is small (about -0.02%). Distributionally, however, the pattern is sharp: poorer households gain, while richer households lose. The gain at the bottom reflects easier insurance via electrification when volatility rises; the loss at the top reflects continued exposure to gas price risk among households that optimally remain on gas given their preferences and equipment.

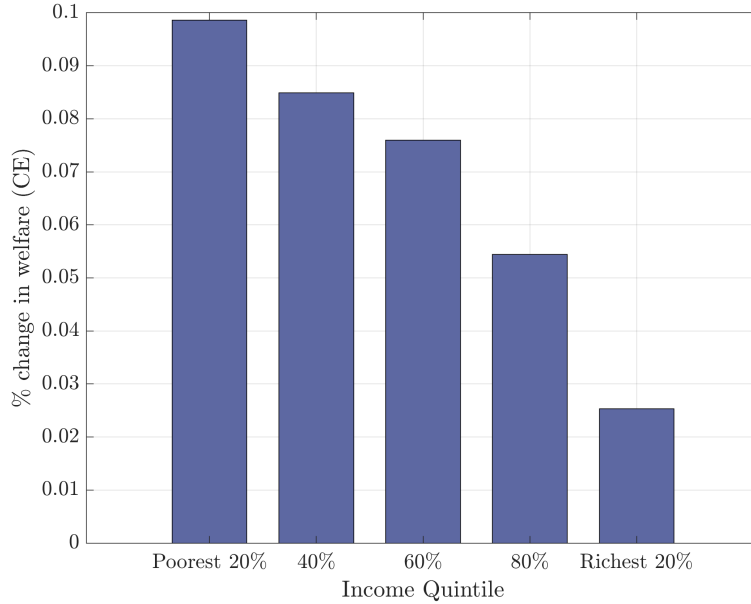


Figure 14: Consumption-equivalent welfare change by income quintile

4.2 Policy implications

The results highlight a distinct volatility channel of electrification: higher gas price volatility induces a shift toward electricity even holding average prices fixed, with stronger extensive-margin responses among poorer households.

Two policy lessons follow. First, short-run stabilization policies that insure households

against gas price spikes (e.g., bill credits, fuel-cost pass-through caps) can attenuate the very risk that drives electrification. Such policies may be desirable on equity grounds, but they trade off near-term protection against slower adoption of the less volatile technology. Designing relief that preserves long-run incentives—e.g., coupling temporary transfers with technology-neutral capital subsidies or heat-pump rebates—can mitigate this tension.

Second, the volatility channel interacts with the clean transition. As more households adopt electricity in response to volatility, demand shifts toward the power sector, which can accelerate learning-by-doing and directed technical change in clean generation and electrification technologies. The distributional incidence is non-neutral: in this model, poorer households do more of the switching. Targeted support for upfront costs (e.g., means-tested rebates, low-interest financing, or on-bill financing) can help ensure that the benefits of lower volatility and cleaner energy are broadly shared while avoiding regressive burdens on households that remain exposed to gas price risk.

Overall, the quantitative message is simple: volatility is not just noise around levels. It reallocates technology choices across the income distribution, modestly raises aggregate consumption, and has distinct distributional and policy consequences.

5 Conclusion

This paper develops a quantitative heterogeneous-agent model to study how households with different income levels respond to energy price fluctuations, particularly those using natural gas. Guided by empirical evidence from the PSID, I document two central facts: first, that lower-income households are more likely to use electricity for heating, while higher-income households rely more on gas; and second, that energy spending is strongly non-homothetic, with poorer households allocating a larger share of their expenditure to energy.

To interpret these patterns and assess their implications, I build a model in which households

choose between electricity and gas under income heterogeneity, incomplete markets, and non-homothetic preferences. Gas prices are subject to stochastic fluctuations, while electricity prices remain stable. The model is calibrated to match key moments from U.S. microdata and replicates well the observed variation in energy use and spending across income groups.

Using the model, I study the distributional effects of energy price risk across two margins. In the short run, households are locked into their energy types. I find that volatility in gas prices is welfare-reducing, especially for low-income households, who are more exposed due to higher energy budget shares. In the long run, however, households can switch energy types. The model predicts that poorer households, facing greater welfare losses from volatility, are more likely to switch to electricity—thereby improving their resilience. As a result, stochastic energy prices generate an endogenous pattern of electrification that is concentrated among the most vulnerable.

Overall, the results underscore the importance of considering non-homothetic preferences and endogenous energy choices when evaluating the distributional consequences of energy market dynamics. They also suggest that volatility—often seen as a purely negative externality—can induce meaningful behavioral adjustment, accelerating transitions toward more stable energy sources.

Future work could incorporate general equilibrium feedbacks, including endogenous energy prices and supply-side constraints, or evaluate the interaction between energy price volatility and targeted policy interventions such as subsidies for electrification or risk-based energy assistance programs. These extensions would allow for a richer understanding of how policy can shape both equity and efficiency in energy transitions.

Appendix 1: Derivations

I derive the optimal consumption bundles given the non-homothetic CES utility function.

The problem, given total expenditure E , is

$$\begin{aligned} \max_{c_c, c_k} \quad & c \\ \text{s.t.} \quad & \theta^{\frac{1}{\mu}} \left(\frac{c_c}{c} \right)^{\frac{\mu-1}{\mu}} + (1-\theta)^{\frac{1}{\mu}} \left(\frac{c_k e_k}{c} \right)^{\frac{\mu-1}{\mu}} = 1 \\ & E = c_c + q_k c_k \end{aligned}$$

The Lagrangian for this problem is:

$$\mathcal{L} = c + \rho \left[1 - \theta^{\frac{1}{\mu}} \left(\frac{c_c}{c} \right)^{\frac{\mu-1}{\mu}} - (1-\theta)^{\frac{1}{\mu}} \left(\frac{c_k e_k}{c} \right)^{\frac{\mu-1}{\mu}} \right] + \lambda [E - c_c - q_k c_k]$$

The first-order conditions with respect to c_c is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_c} = 0 & \implies \theta^{\frac{1}{\mu}} \frac{1-\mu}{\mu} \frac{1}{c} \left(\frac{c_c}{c} \right)^{\frac{-1}{\mu}} = \frac{\lambda}{\rho} \\ & \implies \theta^{\frac{1}{\mu}} \frac{1-\mu}{\mu} \left(\frac{1}{c} \right)^{\frac{\mu-1}{\mu}} \frac{c_c^{\frac{\mu-1}{\mu}}}{c_c} = \frac{\lambda}{\rho} \\ & \implies c_c = \frac{\rho}{\lambda} \theta^{\frac{1}{\mu}} \frac{1-\mu}{\mu} \left(\frac{c_c}{c} \right)^{\frac{\mu-1}{\mu}} \end{aligned}$$

and the first-order condition with respect to c_k is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_k} = 0 & \implies (1-\theta)^{\frac{1}{\mu}} \frac{1-\mu}{\mu} \frac{1}{c} \left(\frac{e_k c_k}{c} \right)^{\frac{-1}{\mu}} = q_k \frac{\lambda}{\rho} \\ & \implies (1-\theta)^{\frac{1}{\mu}} \frac{1-\mu}{\mu} \left(\frac{1}{c} \right)^{\frac{\mu-1}{\mu}} \frac{(e_k c_k)^{\frac{\mu-1}{\mu}}}{e_k c_k} = q_k \frac{\lambda}{\rho} \\ & \implies c_k = \frac{1}{e_k} \frac{1}{q_k} \frac{\rho}{\lambda} (1-\theta)^{\frac{1}{\mu}} \frac{1-\mu}{\mu} \left(\frac{e_k c_k}{c} \right)^{\frac{\mu-1}{\mu}} \end{aligned}$$

Replacing in E above gives:

$$\begin{aligned}
E &= c_c + q_k c_k \\
&= \frac{\rho}{\lambda} \theta^{\frac{1}{\mu}} \frac{1-\mu}{\mu} \left(\frac{c_c}{c} \right)^{\frac{\mu-1}{\mu}} + \frac{1}{e_k} \frac{\rho}{\lambda} (1-\theta)^{\frac{1}{\mu}} \frac{1-\mu}{\mu} \left(\frac{e_k c_k}{c} \right)^{\frac{\mu-1}{\mu}} \\
&= \frac{\rho}{\lambda} \frac{1-\mu}{\mu} \left[\theta^{\frac{1}{\mu}} \left(\frac{c_c}{c} \right)^{\frac{\mu-1}{\mu}} + (1-\theta)^{\frac{1}{\mu}} \frac{1}{e_k} \left(\frac{e_k c_k}{c} \right)^{\frac{\mu-1}{\mu}} \right]
\end{aligned}$$

5.1 First-Order Conditions and Euler Equation

Conditional on the energy type k chosen at the start of the period, the household solves

$$\max_{c, a' \geq -\phi} \frac{c^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}[V(a', z', k; \gamma')] \quad \text{s.t.} \quad E(c, k; \gamma) + a' = (1+r)a + z,$$

where $E(c, k; \gamma)$ is the (non-homothetic CES) *expenditure function* required to attain composite c given prices.

Let λ be the multiplier on the budget constraint. The Lagrangian is

$$\mathcal{L} = \frac{c^{1-\sigma}}{1-\sigma} + \beta \mathbb{E}[V(a', z', k; \gamma')] + \lambda((1+r)a + z - E(c, k; \gamma) - a') + \xi(a' + \phi),$$

with Kuhn–Tucker multiplier $\xi \geq 0$ for the borrowing constraint.

Intratemporal FOC. The first-order condition with respect to c equates marginal utility to the multiplier times the *marginal expenditure* needed to raise c :

$$c^{-\sigma} = \lambda P(c, k; \gamma), \quad \text{where} \quad P(c, k; \gamma) \equiv \frac{\partial E(c, k; \gamma)}{\partial c}. \quad (19)$$

Using the non-homothetic CES aggregator (from the text), the derivative is

$$P(c, k; \gamma) = E(c, k; \gamma)^\mu \left[\theta_c \eta_c c^{(1-\mu)\eta_c-1} + \theta_e \eta_e c^{(1-\mu)\eta_e-1} q_k(\gamma)^{1-\mu} \right], \quad (20)$$

so $P(c, k; \gamma)$ is the shadow “price index” of one more unit of composite consumption c . Intuitively, because preferences are non-homothetic, P depends on the *level* of c (Engel-curve effects), the energy type k , and the current gas shock γ .

Intertemporal FOC (Euler). The FOC with respect to a' and the envelope condition imply

$$-\lambda + \beta \mathbb{E}[V_a(a', z', k; \gamma')] + \xi = 0, \quad V_a(a, z, k; \gamma) = \lambda(1 + r).$$

Combining these gives the standard multiplier recursion

$$\lambda = \beta(1 + r) \mathbb{E}[\lambda'] + \xi,$$

where primes denote next period’s variables. Substituting the intratemporal condition (19) at t and $t+1$ yields the Euler equation in *consumption-price-index* form:

$$\frac{c_t^{-\sigma}}{P(c_t, k_t; \gamma_t)} \geq \beta(1 + r) \mathbb{E}_t \left[\frac{c_{t+1}^{-\sigma}}{P(c_{t+1}, k_{t+1}; \gamma_{t+1})} \right], \quad (21)$$

with equality whenever the borrowing constraint does not bind (i.e., $a' > -\phi$ so $\xi = 0$).

Interpretation. Equation (21) says that the household equates the *marginal utility of one more dollar of expenditure today*, $c_t^{-\sigma}/P_t$, to the discounted expected *marginal utility value of a dollar saved*, $\beta(1 + r)\mathbb{E}[c_{t+1}^{-\sigma}/P_{t+1}]$. The non-homothetic structure enters through the endogenous price index $P(c, k; \gamma)$: because energy is closer to a necessity, P falls less rapidly with c than in a homothetic CES, and depends on which technology k is in place and on the realized gas shock γ . Fuel switching affects the Euler equation only through its impact on $P(c_{t+1}, k_{t+1}; \gamma_{t+1})$ and on the resource constraint (via the fixed cost f in periods when switching occurs); the discrete choice itself does not alter the first-order condition for a' .

Figure 15: Caption

Appendix 2: Additional empirical results

Income and energy consumption

Non-homothetic energy expenditure

The second empirical regularity I document is that energy expenditure exhibits non-homotheticity: the share of total expenditure devoted to energy declines with income. This is a typical characteristic of necessity goods, for which a baseline level of consumption is required regardless of income. Energy use—driven in part by heating, cooking, and basic household functions—fits naturally into this category.

Figure 17 illustrates this pattern by presenting, for each income quintile, two key statistics: average energy expenditure (panel (a)) and average energy expenditure share (panel (b)). The expenditure measure combines annual spending on both gas and electricity. Panel (a) shows that, in absolute terms, wealthier households spend more on energy. For instance, households in the top income quintile spend roughly 1.5 times more than those in the bottom quintile. However, panel (b) reveals a sharply declining trend in the energy expenditure share: while households in the bottom quintile allocate approximately 10% of their total expenditure to energy, this share drops to around 4% for households in the top quintile.

To validate that this pattern is not driven by compositional or confounding factors, I estimate a regression of the log energy expenditure share on log income, controlling for relevant household characteristics and fixed effects. The specification is as follows:

$$\log(\text{energyShare}_{ist}) = \beta_0 + \beta_1 \log(\text{income}_i) + X_{it} + \gamma_t + \gamma_s + \varepsilon_{ist} \quad (22)$$

Here, energyShare_{ist} is the ratio of household i 's energy expenditure to total expenditure

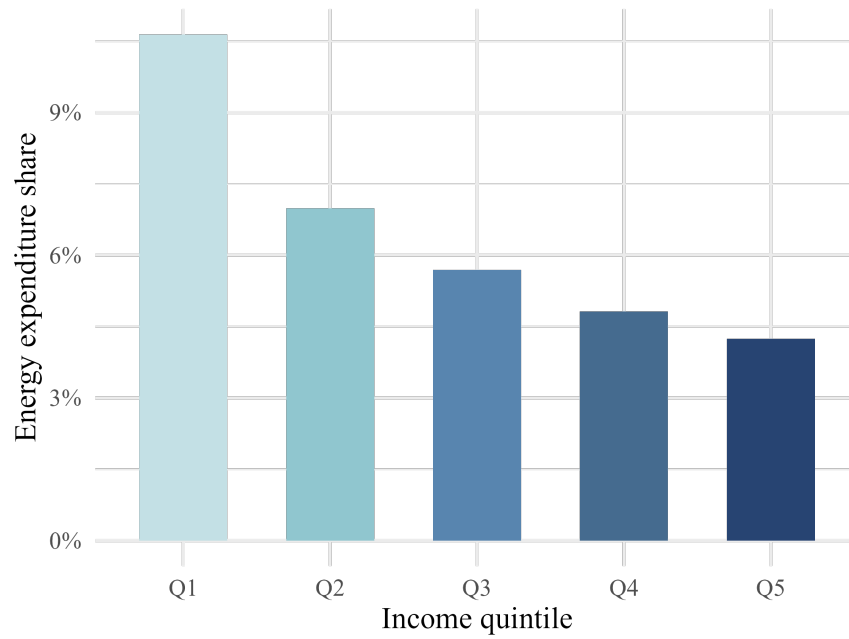


Figure 16: Energy Expenditures Shares by Income Quintile

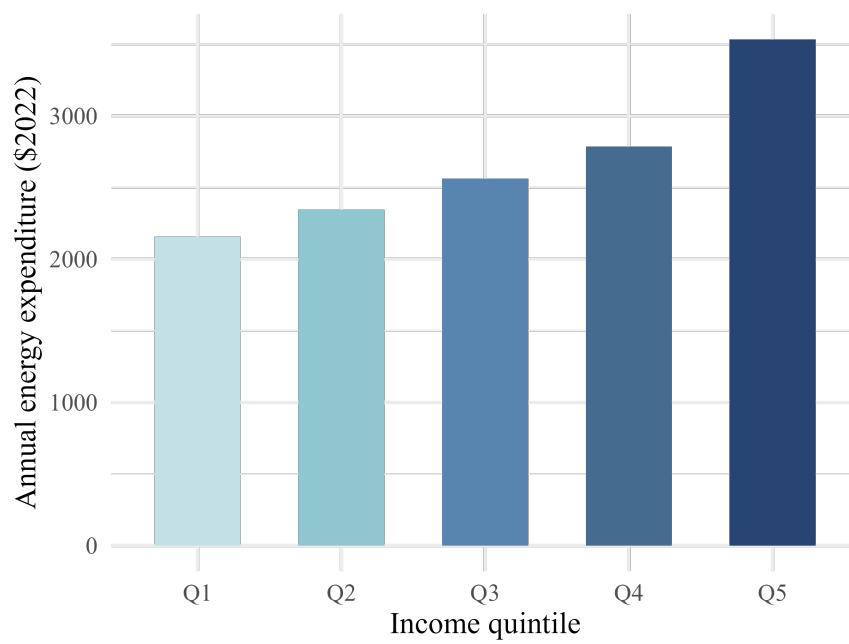


Figure 17: Energy Expenditures by Income Quintile

in state s and year t , and income_i denotes the lifetime income measure described earlier. The control vector X_{it} includes climate controls (heating and cooling degree days), housing characteristics, household size, and demographics, as specified in the previous subsection. The sample is restricted to households using either electricity or gas as their primary heating source.

The coefficient of interest, β_1 , captures the elasticity of energy expenditure share with respect to income. A negative value indicates that higher-income households spend a smaller proportion of their budget on energy, consistent with non-homothetic preferences. Table 5 reports results from three specifications: (1) a baseline model without controls or fixed effects; (2) a specification with controls only; and (3) a fully saturated model including both controls and state and year fixed effects.

In all three models, β_1 is negative and highly significant. In the fully specified model, the estimate implies that a 1% increase in income is associated with an approximate 0.39% reduction in the energy expenditure share. These findings robustly support the view that energy is a necessity good, and that energy consumption is non-homothetic with respect to income.

Table 5: Income and Energy Expenditure Share

	(1) Baseline	(2) + controls	(3) + year/state FE
log income	−0.289*** (0.003)	−0.311*** (0.004)	−0.288*** (0.012)
Controls	No	Yes	Yes
Year FE	No	No	Yes
State FE	No	No	Yes
Observations	41,686	36,764	36,764
R^2	0.15	0.22	0.30

Notes: Coefficients are from linear regressions. Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. The dependent variable is the log of energy expenditure share. Controls and fixed effects are added sequentially across specifications. Controls include heating and cooling degree-days (base 65°F/18°C), age (squared), house tenure, family size (squared) and house type.

Appendix 3: Computational appendix

This appendix describes the solution algorithm, state-space discretization, simulation procedure, and accuracy diagnostics used to solve the model. Code is written in MATLAB; all functions referenced below correspond to routines in the repository.

A3.1. State Space and Discretization

Individual state. The individual state is $s = (a, z, k, \gamma)$: assets a , idiosyncratic labor productivity z , installed energy technology $k \in \{\ell, g\}$ (electricity, gas), and the gas price shock γ .

Grids. Assets use a dense, convex grid with $n_a = 201$ points:

$$a \in [a_{\min}, a_{\max}] = [0, 200], \quad a_j = a_{\min} + (a_{\max} - a_{\min}) \cdot \text{nodeunif}(j)^2,$$

placing more points near the borrowing bound. The technology grid is $k \in \{0, 1\}$ with $0 = \ell$ and $1 = g$.

Shocks. Both z and $\log \gamma$ follow AR(1) processes discretized by the Rouwenhorst method.

$$\log z' = \rho_z \log z + \sigma_z \varepsilon'_z, \quad \log \gamma' = \rho_\gamma \log \gamma + \sigma_\gamma \varepsilon'_\gamma,$$

with $n_y = 5$ nodes for z and $n_\gamma = 2$ nodes for γ . The discrete grids are exponentiated to levels. Transition matrices Q_{yy} and $Q_{\gamma\gamma}$ and stationary distributions (F_y, F_γ) are computed in code via `rouwenhorst` and an eigenvector routine.

Prices and normalization. The non-energy numeraire is 1. Electricity price is constant $q_\ell = \bar{q}_\ell$; gas price is $q_g(\gamma) = \bar{q}_g \cdot \gamma$. Taste-shock locations for switching are (μ_ℓ, μ_g) ; dispersion is λ .

A3.2. Preferences and Within-Period Problem

Period utility is $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$ (log for $\sigma = 1$). The non-homothetic CES structure implies an *expenditure function* $E(c, k; \gamma)$ mapping composite consumption c into the minimum dollars required at prices $(1, q_k(\gamma))$:

$$E(c, k; \gamma) = \left[(1 - \theta) c^{1-\mu} + \theta q_k(\gamma)^{1-\mu} c^{\eta(1-\mu)} \right]^{\frac{1}{1-\mu}}.$$

Given c and $q_k(\gamma)$, the code computes

$$\text{non-energy } c_c, \quad \text{energy } e, \quad \text{and } a' = (1 + r)a + z - c_c - q_k(\gamma)e,$$

using closed-form demand shares from the aggregator (see `savings_p` and `savingsss_p`).

When switching, a fixed financial cost f is subtracted from resources.

5.2 Dynamic Programming Problem and Bellman Operators

For a given k , the value of keeping k is

$$V^n(s) = \max_{c, a' \geq \phi} \left\{ u(c) + \beta \mathbb{E}[V(a', z', k; \gamma')] \right\} \quad \text{s.t.} \quad E(c, k; \gamma) + a' = (1 + r)a + z.$$

The value of switching to $k' = 1 - k$ is analogous, with (i) k' used in prices/aggregator, (ii) the fixed cost f deducted from the budget, and (iii) the *taste shock* added to the continuation value. Because taste shocks are Type-I Extreme Value with dispersion λ , the ex-ante value function combines the two discrete alternatives using the log-sum formula:

$$V(s) = \lambda \log \left(e^{V^n(s)/\lambda} + e^{(V^s(s) + \nu(s))/\lambda} \right),$$

where $\nu(s)$ is the state-dependent taste location (gas vs. electricity). In the code, V^s is first computed and then shifted by ν ; the overall V is computed via a *log-sum-exp* step that improves numerical stability by subtracting $\max\{V^n, V^s\}$ before exponentiating (see `solve_dynamic_p.m`).

5.3 Solution Algorithm

Interpolation and expectations. Given a guess $V^{(iter)}$ on the full tensor grid, the conditional expectation is computed by multiplying $V^{(iter)}$ by the Kronecker product of transition matrices, then wrapping a `griddedInterpolant` object (spline in state, linear extrapolation) to evaluate V off-grid inside the one-dimensional inner optimization.

Inner (within-period) maximization. For each state, the algorithm maximizes over the scalar c subject to the budget and the borrowing constraint. Feasible bounds for c are computed by bisection on the implied savings rule (functions `savings_p`, `savingsss_p`). Given state s , the routine:

1. computes feasible $[c_{\min}, c_{\max}]$ with bisection (absolute tolerance 10^{-14});
2. solves the scalar maximization via golden-section search (`solve_golden`) for both “keep” and “switch” branches;
3. evaluates V^n and V^s with the envelope/continuation operator (`valfunc_p`, `valfuncs_p`).

Infeasible choices that violate $a' \geq \phi$ are penalized by a large negative utility (-10^{10}), which effectively imposes the constraint.

Value function iteration with policy improvement. The outer loop updates V using the log-sum step. Every 15 iterations, a Howard-style improvement is applied: holding the current policies (c^n, c^s) fixed, the code iterates the value operator 15 additional times to accelerate convergence. The stopping criterion is $\|V^{(iter)} - V^{(iter-1)}\|_2 < 10^{-2}$ or a hard cap of 500 iterations.

Switching probabilities. Given V^n and $V^s + \nu$, the model-implied probability of switching is the logit formula

$$\Pr(\text{switch} \mid s) = \frac{\exp\left((V^s(s) + \nu(s))/\lambda\right)}{\exp(V^n(s)/\lambda) + \exp\left((V^s(s) + \nu(s))/\lambda\right)}.$$

This object is stored as `pfs.sw` and later used in simulation.

5.4 Simulation and Stationary Distribution

To compute ergodic moments, I simulate $N = 10,000$ households for $T = 5,000$ periods with a burn-in of 2,000 periods (`solve_ergodic_p`). Income paths z are idiosyncratic; the gas shock γ is simulated as a common aggregate process for all households (consistent with the model’s interpretation). Policy functions are evaluated via `griddedInterpolant` objects over (a, z, k, γ) (linear interpolation / nearest extrapolation). Each period:

1. draw a Bernoulli switch with probability `sw`;

2. set composite consumption C , energy/non-energy spending (e, c_e) , and next assets a' from the appropriate (keep/switch) policy;
3. update k' , a' ; advance z', γ' using $Q_{yy}, Q_{\gamma\gamma}$.

Post burn-in, I compute aggregates and targeted moments (wealth-to-income, energy shares, switching frequency, etc.) as cross-sectional averages.

5.5 Parameters Used in Solution

Assigned (calibrated externally) parameters:

$$\sigma = 1, \quad r = 0.03, \quad \phi = 0, \quad \mu = 0.275, \quad \rho_\gamma = 0.8, \quad \sigma_\gamma = 0.4, \quad \rho_z = 0.92, \quad \sigma_z = 0.285, \quad \theta = 0.07.$$

Estimated (via SMM): $\beta = 0.95618$, $\eta = 0.04$, $(\bar{q}_g, \bar{q}_\ell)$, (μ_ℓ, μ_g) , f , and λ (see Table 2 in the main text).

Computational controls:

$$n_a = 201, \quad n_y = 5, \quad n_k = 2, \quad n_\gamma = 2, \quad \text{tol}_{\text{VFI}} = 10^{-2}, \quad \text{tol}_{\text{bisect}} = 10^{-14},$$

Howard improvement every 15 iterations, 15 inner Howard steps, and up to 500 VFI iterations.

5.6 Accuracy and Diagnostics

I report three standard diagnostics:

(i) *Euler-equation errors*. For a random subset of states, I compute the Euler residual in the “price-of-utility” form

$$\frac{c_t^{-\sigma}}{P(c_t, k_t; \gamma_t)} - \beta(1+r) \mathbb{E}_t \left[\frac{c_{t+1}^{-\sigma}}{P(c_{t+1}, k_{t+1}; \gamma_{t+1})} \right],$$

where $P(c, k; \gamma) = \partial E / \partial c$. Median absolute errors are small (reported in the online appendix).

(ii) *Policy monotonicity.* Savings are (weakly) increasing in a and z on the grid; switching probabilities are increasing in $(V^s + \nu) - V^n$ by construction.

(iii) *Grid, interpolation, and tolerance robustness.* Doubling n_a and n_y or tightening tol leaves targeted moments and key comparative statics unchanged at two significant digits. Using linear instead of spline interpolation for V yields indistinguishable aggregates and slightly slower convergence.

5.7 Algorithm Pseudocode

[H] [1] Build grids for (a, z, k, γ) ; construct $Q_{yy}, Q_{\gamma\gamma}$ via Rouwenhorst. Initialize $V^{(0)}(s) = 1$ for all states. $iter = 1, \dots, 500$ Form interpolant V^{int} from $V^{(iter-1)}$. each state s Compute feasible $[c_{\min}, c_{\max}]$ (bisection on savings) for keep and switch. $c^n = \arg \max_c \text{valfunc_p}(c, s)$ via golden search; get $V^n(s)$. $c^s = \arg \max_c \text{valfuncs_p}(c, s)$ via golden search; get $V^s(s) + \nu(s)$. Update V via log-sum-exp with scale λ . $iter \bmod 15 = 0$ Apply 15-step Howard improvement holding (c^n, c^s) fixed. Stop if $\|V^{(iter)} - V^{(iter-1)}\|_2 < 10^{-2}$. Recover policies (a', C, e, c_c) and switching probability $\Pr(\text{switch} \mid s)$. Simulate $N \times T$ panel; drop burn-in; compute moments.

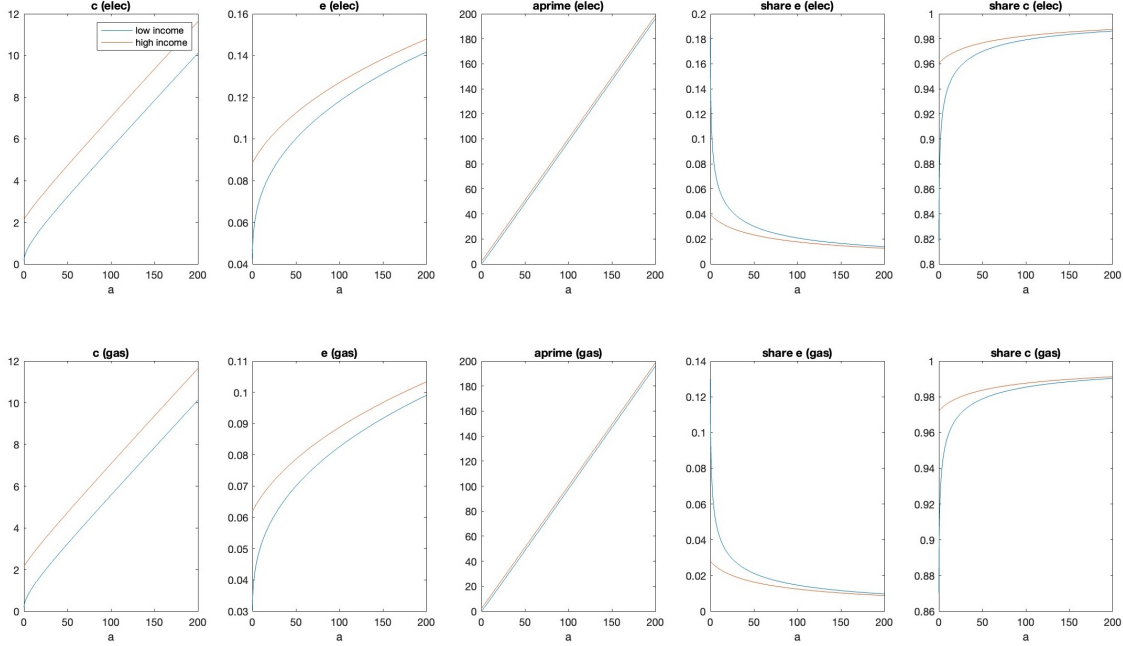


Figure 18: Optimal policy functions

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